Bihar Engineering University, Patna

B.Tech. 7th Semester Examination, 2023

Course: B.Tech. ***Time: 03 Hours***

Code: 100708 ***Subject: Biology for Engineers*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **SIX** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer of the following (Any seven question only): **[2 x 7 = 14]**

(a) Organism that prepare their own food are called as

(i) Autotrophs (ii) Heterotrophs

(iii)Consumers (iv) Producers

(b) Louis Pasteur discovered_____

(i) Microscope (ii) Sterilization

(iii) Evolution (iv) Pasteurization

(c) The Digestive system of cell is known as

(i) Chloroplast (ii) Mitochondria

(iii) Cell wall (iv) Lysosomes

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(d) Amoeba is an example of _____

(i) Multicellular (ii) Neutrophylls

(iii) Unicellular (iv) Extremophiles

(e) The first form of life on earth was

(i) Animal (ii) Plant

(iii) Zooplankton (iv) Microorganism

(f) The simplest amino acid is

(i) glycine. (ii) alanine

(iii) asparagine. (iv) tyrosine

(g) The branch of biology dealing with the study of cells is known as

(i) Histology (ii) Cytology

(iii) Psychology (iv) Physiology

(h) In a plant, the xylem tissue conducts

(i) Water (ii) Food

(iii)Water and food (iv) None of these

(i) The red colour of human blood is due to

(i) Myoglobin (ii) Haemoglobin

(iii)Immunoglobulin (iv)Haptoglobin

(j) The suicidal bags of the cell are

(i) Lysosomes (ii) Ribosomes

(iii) Dictyosomes (iv) Phagosomes

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Q.2	(a)	What do you understand by **Brownian Motion**? Explain with suitable examples.	**[7]**
	(b)	Differentiate between endothermic reaction and exothermic reaction.	**[7]**
Q.3	(a)	What do you understand by **Hierarchy of Biological Classification**?	**[7]**
	(b)	Describe photosynthesis for the production of energy.	**[7]**
Q.4	(a)	Describe the Gene Mapping and Gene Interaction with suitable example.	**[7]**
	(b)	Write an essay on the Importance of **Biology for engineers**.	**[7]**
Q.5	(a)	Explain single celled organism and state its process of reproduction.	**[7]**
	(b)	What do you understand by **Monomers**? Differentiate monomer with polymers with suitable example	**[7]**
Q.6	(a)	Write short note on **SUGAR** and **NUCLEOTIDES**.	**[4]**
	(b)	What do you understand by **Ecosystem**? Write short notes on its structure.	**[10]**
Q.7	(a)	What do you mean by Respiration? Explain the types of Respiration.	**[7]**
	(b)	Describe the mechanism of mitosis cell division.	**[7]**
Q.8		Draw a well-labelled diagram of animal and plant cell and list specific functions of each cell organ.	**[14]**
Q.9	(a)	Write in brief properties and examples of all three major kingdoms of life.	**[7]**
	(b)	What is glycolysis? Draw the flow diagram of different stages of glycolysis showing preparatory and payoff phase.	**[7]**

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